

Date	15 July 2026
Project Name	Smart Lender AI: An Intelligent Machine Learning System for Loan Approval Prediction
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Maximum Marks	2 Marks

DEFINE PROBLEM STATEMENTS

Formulating logical, customer-centric problem statements is crucial for translating operational inefficiencies into technical requirements. The traditional credit underwriting workflow is notoriously prone to manual evaluation bottlenecks, inconsistent risk-tolerance assessment, and elevated default rates on outstanding assets. To address these multi-layered challenges, we apply a design-thinking paradigm from the perspective of various stakeholders inside the banking system.

Customer Problem Statement Template

Create a problem statement to understand your customer's point of view. This template helps you focus on what matters to create experiences people will love. A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face.

PS No.	I am... (The Customer)	I'm trying to... (The Goal)	But... (The Obstacle)	Because... (The Root Cause)	Which makes me feel... (The Emotion)
PS-1	A Retail Loan Applicant	Apply for a home or personal loan to secure immediate financing.	The processing cycle takes weeks and lacks clear milestone status updates.	Bank staff rely on physical document vetting, manual calculations, and slow multi-tier approval paths.	Anxious, frustrated, and inclined to look for alternative FinTech lenders.
PS-2	A Bank Credit Risk Officer	Evaluate and classify incoming loan applications to protect capital liquidity.	Assessing multi-dimensional profiles (income ratios, credit ratings, geographic fields) takes too long and introduces human error.	There is no centralized intelligent platform to instantly analyze non-linear dependencies across demographic and financial inputs.	Overwhelmed by backlogs, leading to inconsistent application approvals.
PS-3	The Chief Risk Officer (CRO)	Maintain a low non-performing asset (NPA) ratio and improve overall portfolio yield.	A high volume of risky applications are approved, leading to loan defaults.	Traditional scorecard rules use primitive binary boundaries that cannot capture hidden patterns or multi-feature correlations.	Deeply concerned about operational exposure, bad debt, and regulatory audits.

System-Level Problem Synthesis

An analysis of these stakeholder challenges points to a single core issue: **the lack of an automated, data-driven, and highly reliable decision system**. By implementing **Smart Lender AI**, the manual underwriting workflow is replaced with an intelligent machine learning pipeline.

This automated pipeline cleans incoming applicant data, applies robust statistical transformations, and leverages high-performing classification models (such as XGBoost and Random Forest) to produce immediate eligibility assessments. This technical approach directly addresses stakeholder pain points—providing instant processing for applicants, reliable analytical support for credit officers, and minimized default rates to protect bank assets.